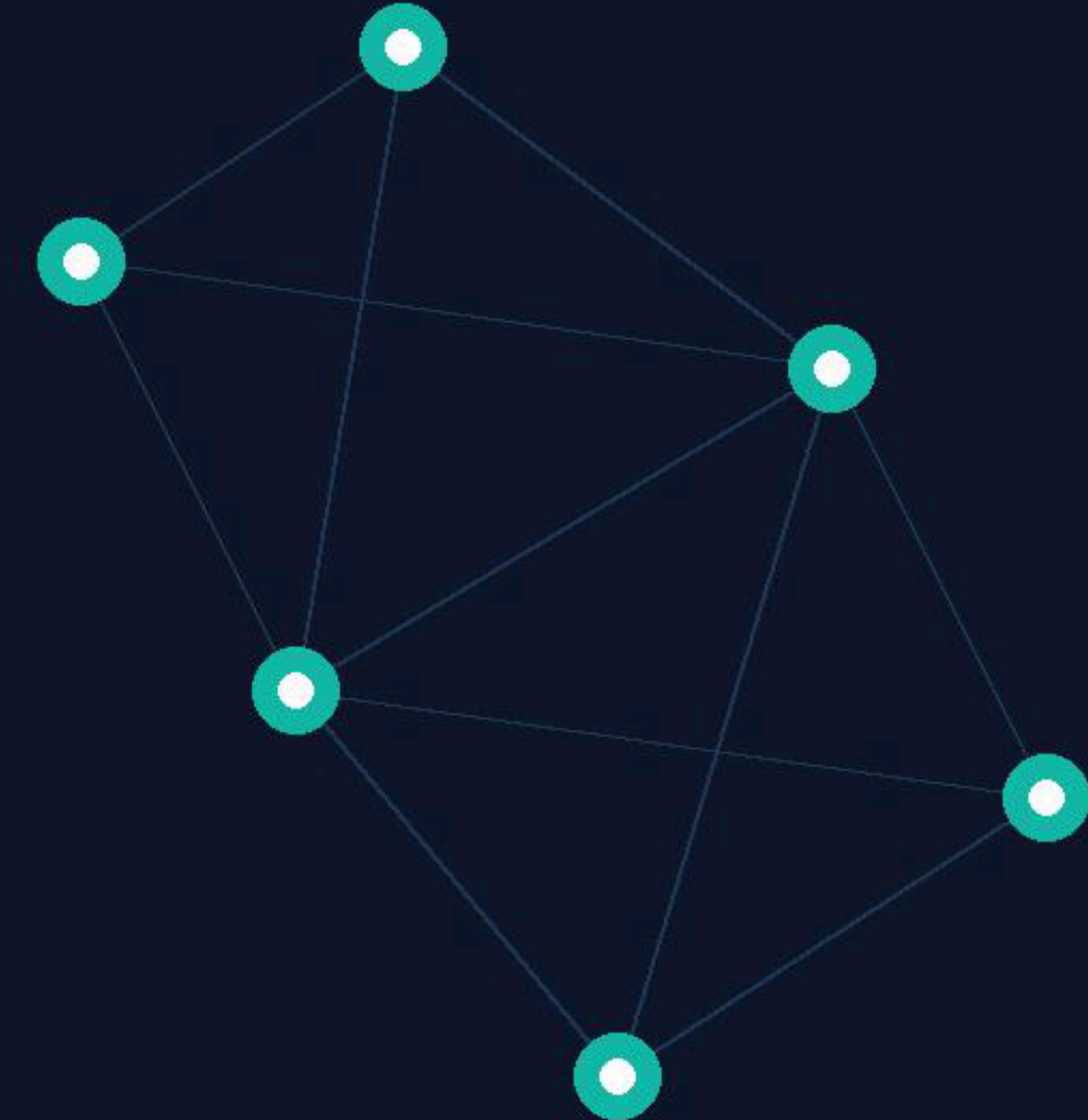


Smart Asset Management and Resource Allocation

Platform Design Document

Project Team Members:

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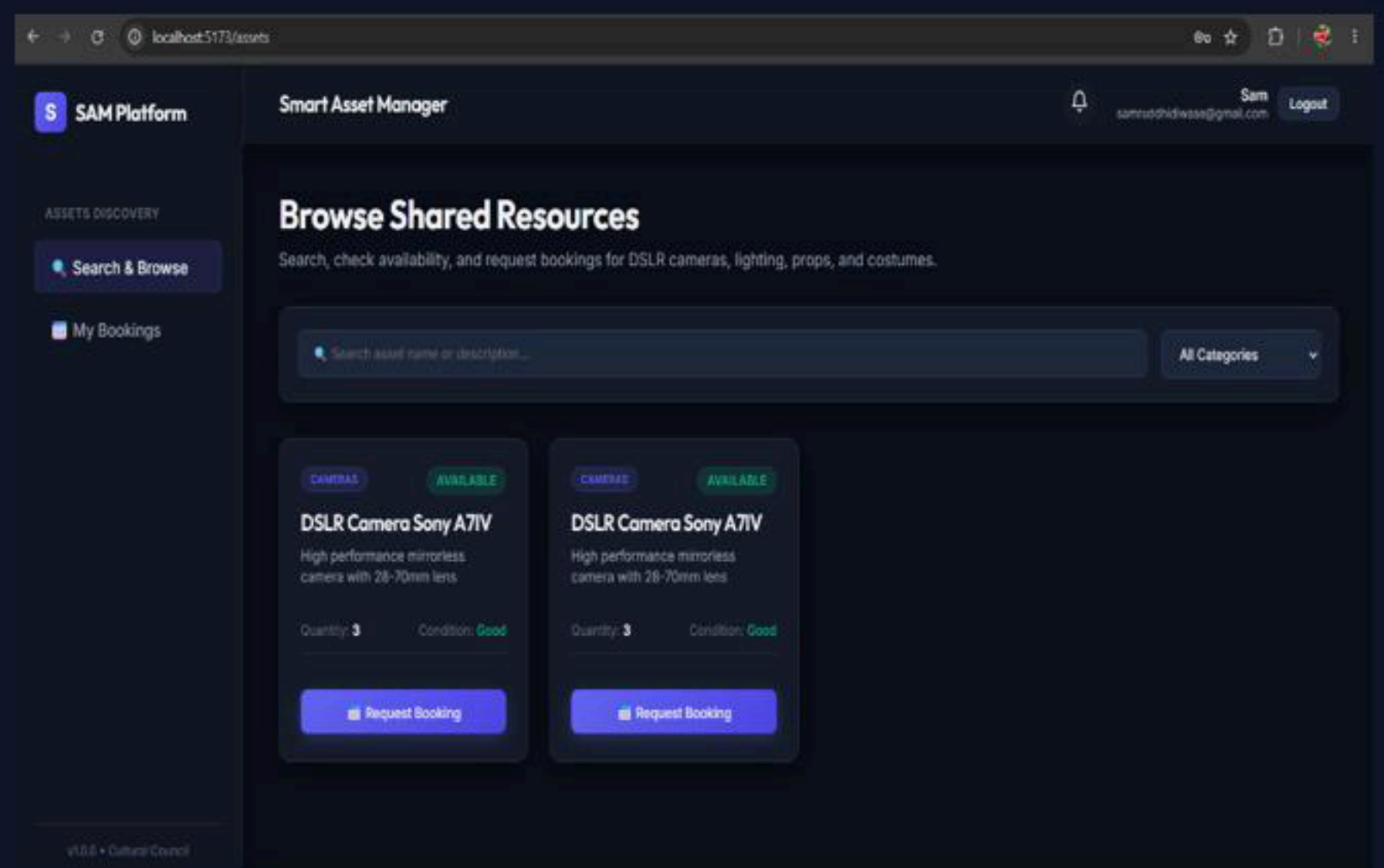


Problem Understanding & Pain Points

Existing Challenges in Asset Tracking

- **Manual Management**
Relying on physical logs and spreadsheets prone to human error.
- **Resource Underutilization**
Assets sit idle due to lack of scheduling visibility and awareness.
- **Asset Loss & Damage**
No custodian records, making it impossible to assign accountability.
- **No Real-Time Visibility**
Borrowers cannot check if an item is available, leading to overlaps.
- **Inefficient Approvals**
Coordinators waste time manually verifying resource availability.

THE SAM SOLUTION & OTP SECURITY

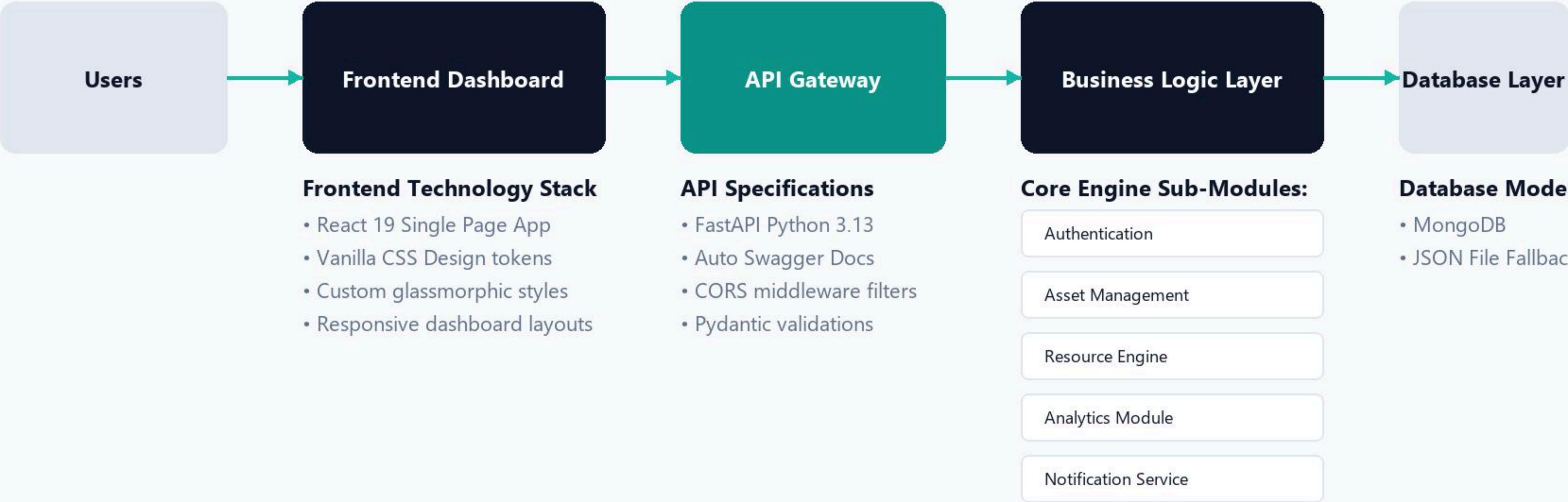


The screenshot displays the SAM Platform interface. The header includes the 'SAM Platform' logo and 'Smart Asset Manager' title. The main content area is titled 'Browse Shared Resources' and features a search bar and a list of assets. Two identical asset cards are shown, each for a 'DSLR Camera Sony A7IV' with a quantity of 3 and a condition of 'Good'. Each card has a 'Request Booking' button. The footer of the interface shows 'v1.0.0 • Cultural Council'.

- Dynamic OTP: Security code prints only to backend console terminal logs.
- First Registrant: Automatically granted full system administrator role.
- Admin Access: admin@sam.iitr.ac.in | Password: AdminPassword123
- Borrower Access: consumer@sam.iitr.ac.in | Password: UserPassword123
- Role Separation: Limits inventory control/approvals strictly to Admin users.

Decoupled Service System Architecture

The platform adopts a modular micro-service style architecture separating interface views from backend business rules.



Data Model & Table Architecture

Users Table	
user_id	: String (PK)
name	: String
email	: String (Unique)
role	: String ('admin' 'user')

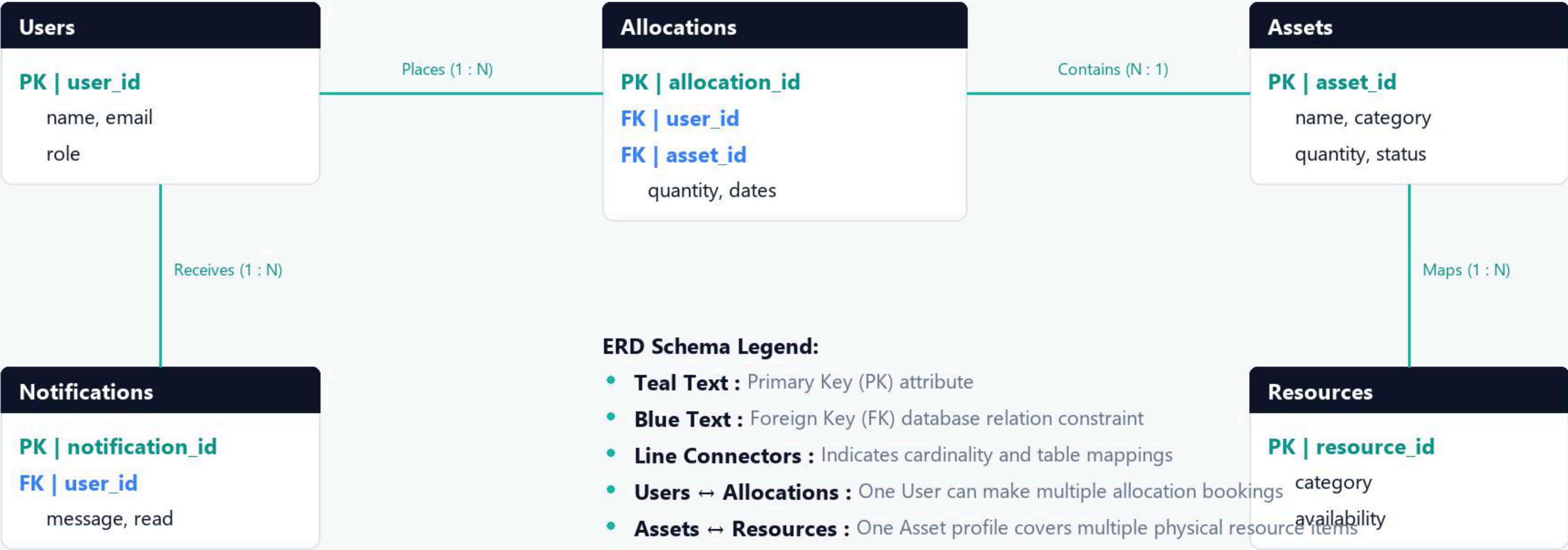
Assets Table	
asset_id	: String (PK)
asset_name	: String
category	: String
quantity	: Integer
status	: String ('Available' 'Repair')

Resources Table	
resource_id	: String (PK)
category	: String
availability	: Boolean

Allocations Table	
allocation_id	: String (PK)
user_id	: String (FK -> Users.user_id)
asset_id	: String (FK -> Assets.asset_id)
quantity	: Integer
start_date	: ISO Date String
end_date	: ISO Date String
status	: String ('pending' 'issued')

Notifications Table	
notification_id	: String (PK)
user_id	: String (FK -> Users.user_id)
message	: String
is_read	: Boolean
timestamp	: ISO Date String

Entity Relationship Diagram (ERD)



REST API Endpoint Specifications

METHOD	ENDPOINT PATH	MODULE SECTION	DESCRIPTION / SPECIFICATIONS
POST	/auth/register	Authentication	Register new profile. First registrant becomes Admin.
POST	/auth/login	Authentication	Authenticates password and returns a session JWT.
GET	/assets	Inventory	Fetches list of all registered council equipment.
POST	/assets	Inventory	Registers a new equipment profile. [Admin Only]
PUT	/assets/{id}	Inventory	Updates equipment info and specifications. [Admin]
DELETE	/assets/{id}	Inventory	Removes asset profile from the catalogs. [Admin]
POST	/bookings	Allocations	Requests bookings, checks date overlaps.
GET	/bookings/my	Allocations	Fetches request history specific to borrower.
GET	/dashboard/stats	Analytics	Aggregates utilization indexes and trends. [Admin]

Architectural Decisions & Catalog UI

Modular Architecture

Decoupled React frontend and FastAPI backend enables independent scaling.

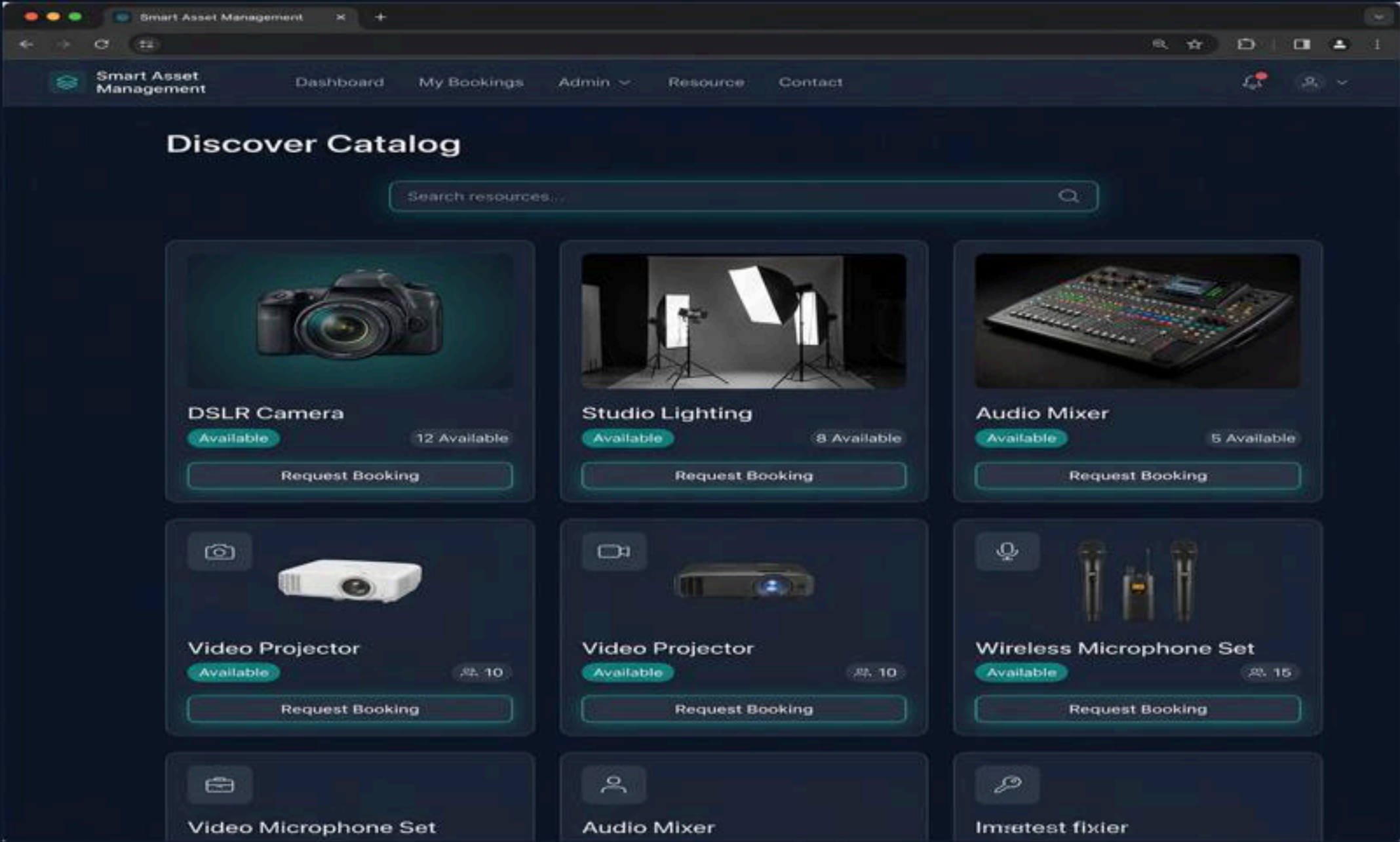
Fail-Safe Database Fallback

Auto-detects database status; degrades gracefully to local JSON fallback database.

Role-Based Access Control

Segregates Admin controls (approvals, CRUD, logs) from borrower options.

PRODUCTION DISCOVER & BOOKING CATALOG



- Search-as-you-type filter bars automatically query catalog databases.
- Media Manager: Admins upload image files or capture live photos via device camera.
- QR Operations: Admin generates equipment labels for instant transaction scans.

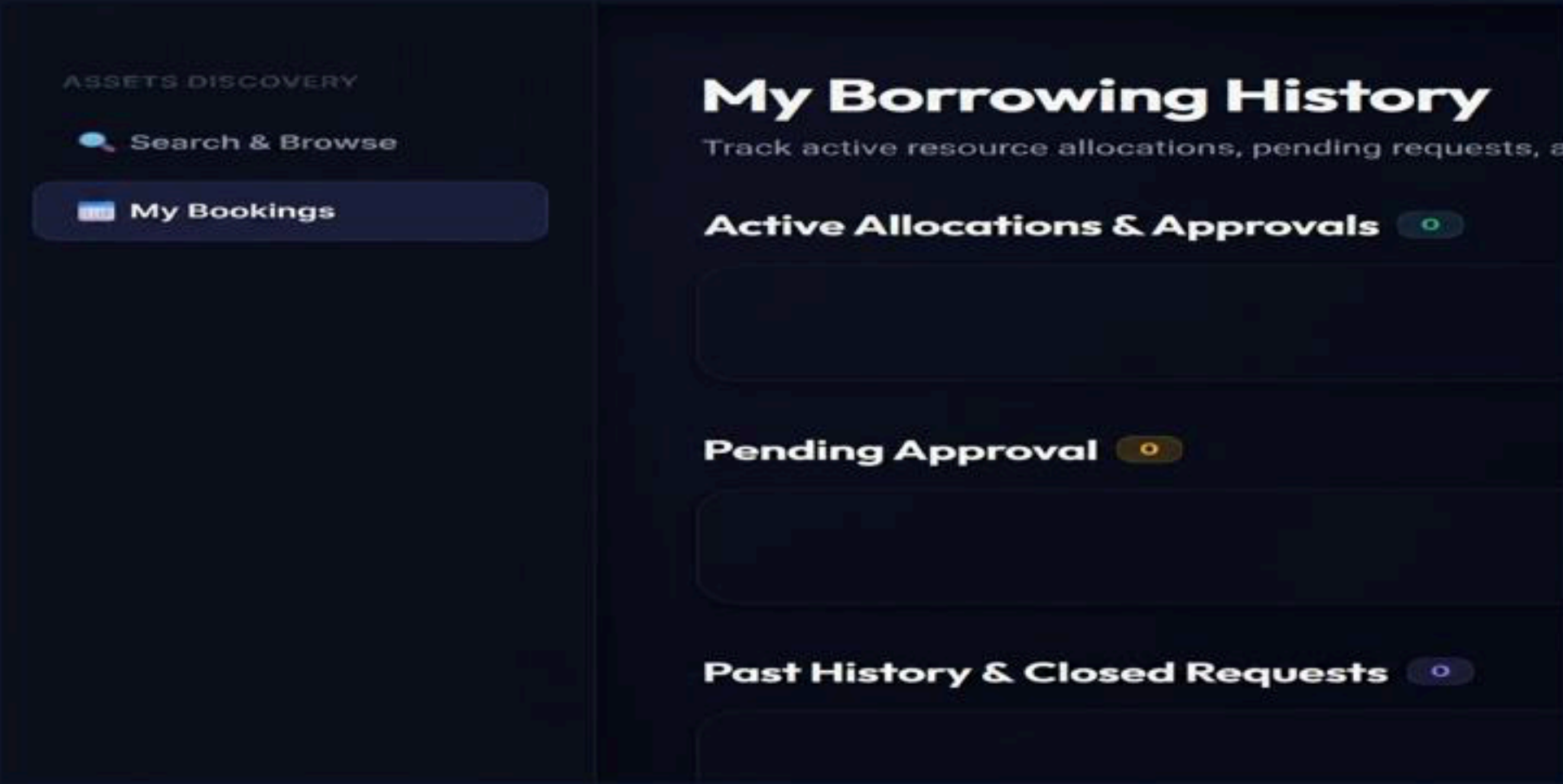
Key Operational Outcomes

- **Improved Asset Visibility**
Central catalog with real-time status reduces request friction.
- **Efficient Utilization**
Date overlap calculations prevent duplicate bookings.
- **Reduced Manual Effort**
Replaces paper logs with simulated barcode scans.

Future Development Roadmap

- **AI-Based Allocation Options**
Smart recommendations based on historical usage logs.
- **Predictive Health Monitoring**
Flags maintenance checks based on checkout frequencies.
- **IoT GPS / RFID Tracking**
Integrating physical sensor checks for security.

Borrower Transaction & Booking Queue:



THANK YOU

IIT Roorkee Cultural Council

Smart Asset Management Platform